Project 2: Write-Up

Course: STATGR5243_001_2025_3 – Applied Data Science
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1 Approach description

The goal is to predict where Dictyostelium cells will aggregate using only the first few movie frames. The models implemented take K early frames and predict a heatmap of likely aggregation locations. The predicted center is the brightest point in this heatmap.

2 Methods

2.1 Data description

The dataset consists a time-lapsed movie stored in a Zarr file as 5D arrays with shape (T, C, Z, H, W) = (34, 1, 32, 256, 256), where:

- T: Timepoints
- C: Imaging Channels
- Z: Slices per Timepoint
- H: Height (pixels)
- W: Width (pixels)

For this project, a single imaging channel was used. The Z-slices were collapsed using max-intensity projection that produced a 3D array (T, H, W). No manual masks or annotations were provided as the final frame itself gives the aggregation pattern used to define ground truth. A Gaussian heatmap centered on the centroid of the last frame serves as the target for all the training.

2.2 Preprocessing

After loading the 5D Zarr file, a max-intensity projection is applied across the Z dimension to obtain 2D frames. The single imaging channel is then selected and all frames are converted to `float32`. Short sequences of K consecutive frames (a temporal window) are created so that each sample contains K = 4 frames as input. The sequence data are split into training, validation and test sets using a 70/15/15 ratio respectively.

2.3 Models

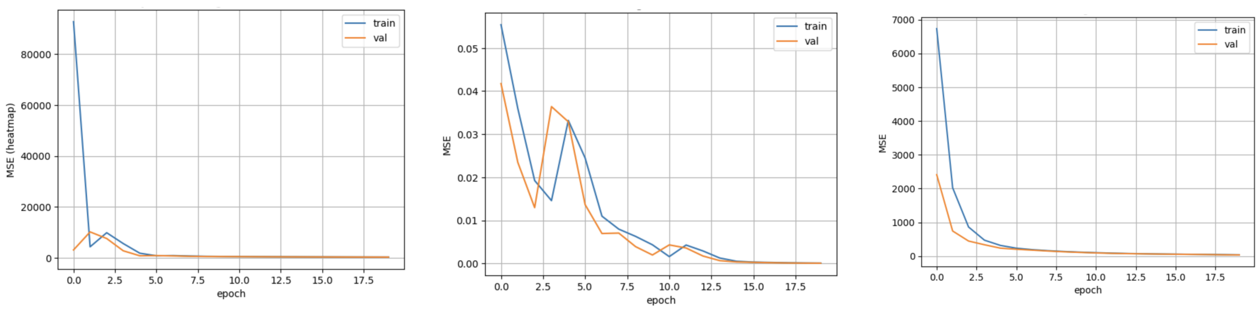
Three models were trained and their performance compared. All of them used mean squared error (MSE) between the predicted and target heatmaps, the Adam optimizer, a learning rate of 1×10^{-3} and 20 training epochs.

- **TinyCNN *(baseline)***: A small 3D CNN that processes all K = 4 frames at once using a temporal 3D convolution, followed by 2D convolutions to produce a heatmap. In addition to the shared settings above, this model also uses weight decay of 1×10^{-5}.

- **LSTM**: Each frame is passed through a shallow CNN (using padded 3×3 convolutions and ReLU activations) to extract spatial features. These features are then passed through the LSTM which learns temporal patterns across K = 4 input frames. The final hidden state is mapped to a heatmap showing the predicted final aggregation center.
- **Small U-Net**: The model takes a single frame as input and again predicts a heatmap showing the final aggregation center. The encoder uses two convolution layers with ReLU and max-pooling to extract features. The bottleneck processes these features before the decoder upsamples them back to the original size. Skip connections link encoder and decoder features which help the model preserve finer spatial details.

3 Results

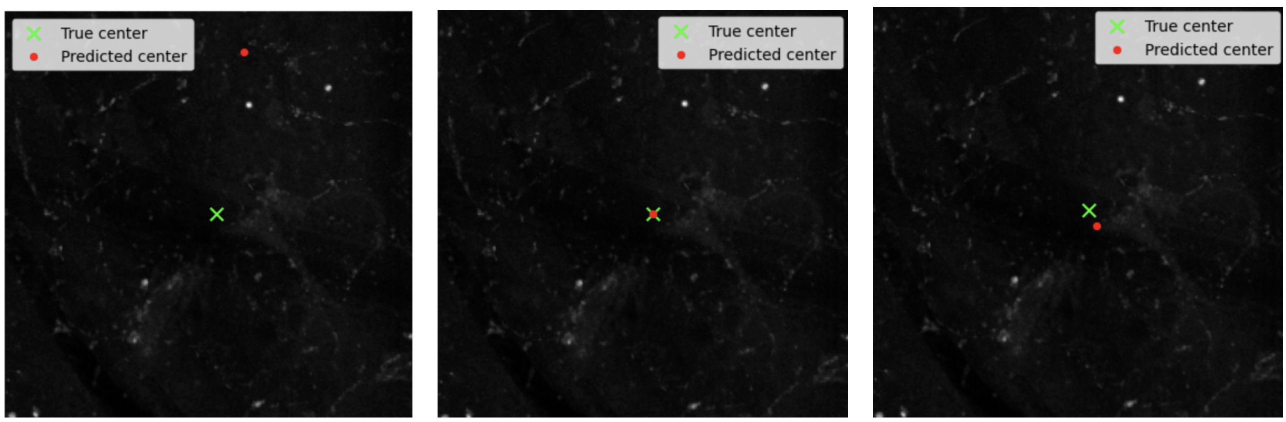
Figure 1 show the training and validation loss. In Figure 1b one can see that the training and validation error overshoots, this is normal early traning behavior and happens when the model is still adjusting its weights.



(a) TinyCNN (b) LSTM (c) Small U-Net

Figure 1: Training and Validation Loss for All Three Models

The LSTM performed by far the best (0.36 px / 0.87 µm), with an AUROC of 1.000 and an average precision of 0.992. The baseline TinyCNN performed poorly, while the Small U-Net did noticeably better than TinyCNN. All metrics are shown in Table 1. For a visual comparison, see Figure 2, where the true center is shown together with each model's predicted center.



(a) TinyCNN (b) LSTM (c) Small U-Net

Figure 2: Early Frames with Predicted Hotspot and center Overlayed for All Three Models

For the best-performing model, LSTM, an analysis was performed to measure how its error changes when only the first N frames are available. For each N, all later frames are zeroed out and the center error is computed.

Model	Center error (px)	Center error (μm)	AUROC	Average Precision
TinyCNN	187.00 ± 101.81	450.68 ± 245.36	0.481 ± 0.070	0.002 ± 0.000
LSTM	0.36 ± 0.00	0.87 ± 0.00	1.000 ± 0.000	0.992 ± 0.000
Small U-Net	11.63 ± 1.19	28.02 ± 2.87	0.527 ± 0.008	0.002 ± 0.000

Table 1: Model comparison with mean $\pm$ 95% CI for center error and heatmap metrics.

Repeating this for all N shows how quickly the model improves as more frames are provided, see Figure 3. The LSTM becomes fairly accurate by frame 3, and by frame 4 it predicts the aggregation center correctly.

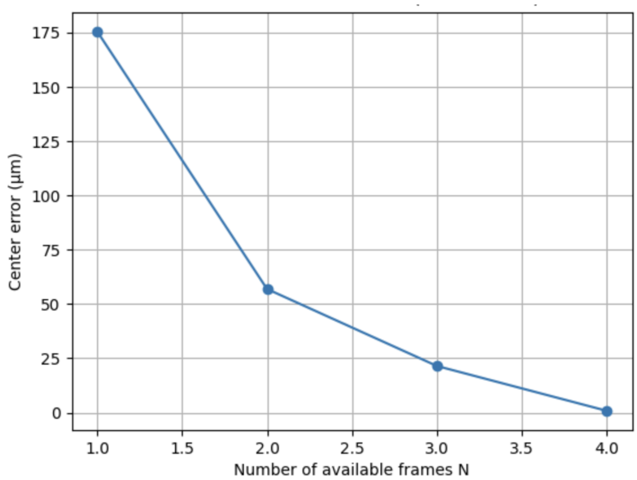


Figure 3: Center error of the LSTM model as a function of the number of available frames.

Lastly, an analysis was performed that tests how well each model handles lower-resolution input. The images were downsampled and then upsampled again, and the center error measured. TinyCNN degrades heavily at lower resolutions, the Small U-Net stays mostly stable and the LSTM remains fully robust with no loss in accuracy.

	Resolution scale		
Model	**1.00**	**0.50**	**0.25**
TinyCNN (µm)	468	2826	1473
TinyCNN drop (%)	0.0	504	215
LSTM (µm)	0.87	0.87	0.87
LSTM drop (%)	0.0	0.0	0.0
Small U-Net (µm)	28.8	26.5	27.4
Small U-Net drop (%)	0.0	-7.9	-5.0

Table 2: Resolution robustness showing the error (µm) and relative drop at different input resolutions.

4 Conclusion

The LSTM clearly outperformed the purely spatial models TinyCNN and Small U-Net. This shows that the temporal component of the LSTM is crucial, allowing it to predict the true center accurately and remain robust across different resolutions.

5 Acknowledgments

The author used OpenAI's ChatGPT (2025) to improve readability and support code development. All analysis and interpretations are the author's own.